

Identity Drift Benchmark — Research 4

Does TTS voice identity drift over 5-minute synthesized monologues?

Date	2026-05-25
GPU	NVIDIA RTX 3060 Laptop (6GB VRAM)
Python	3.12 (Chatterbox, Kokoro), 3.10 (XTTS-v2)
Reference	neutral_p229_002.wav (VCTK Corpus 0.92)
Models	3: Chatterbox 0.1.7, XTTS-v2 0.22.0, Kokoro 0.9.4
Total audio	60 windows across 3 × ~5min files
Primary metric	IQR-scaled Euclidean distance in 7-feature acoustic space
Gate status	PASSED

Methodology

1. Reference

p229 (female, Southern England), 3.88 s, f0 = 192.7 Hz, CPP = 13.36 — same reference used across Papers 2–4.

2. Models

Model	Cloning	Parameters	Sample Rate
Chatterbox 0.1.7	Yes (audio_prompt_path)	exaggeration=0.3	24 kHz
XTTS-v2 0.22.0	Yes (speaker_wav)	language=en	22 kHz
Kokoro 0.9.4	No (voice=af_heart)	N/A	24 kHz

3. Generation

868-word narrative text. Chatterbox chunked into 10 × ~100 words (CUDA context limit). XTTS and Kokoro generated whole. Seed = 42. Target ~5 min each.

4. Windowing

15-second non-overlapping windows. 60 total windows (chatterbox=19, xtts=21, kokoro=20).

5. Features (extracted per window)

f0_mean, f0_std, cpp_mean (Praat CPPS, copied from Paper 2), spectral_flatness, spectral_tilt_ratio, mfcc_mean, rms_mean, spectral_centroid. All resampled to 16 kHz.

6. Voiced-Frame Filtering

Energy threshold + librosa.pyin voiced_probability ≥ 0.50, yin fallback.

7. Gate

Reference voiced ratio ≥ 10%, ≥ 0.2 s voiced, no clipping. Per-model: ≥ 50% windows pass quality check.

8. Scoring

IQR-based robust scaling pooled across all valid windows. Per-window drift = mean of |feature_value – reference_value| / scale. drift_increase = drift_late – drift_early (first/last 5 windows). drift_slope = scipy.stats.linregress across all windows.

Gate Results

Gate status: PASSED

All audio passed quality checks. Reference and all 60 generation windows are valid for analysis.

Reference Gate

Path	Duration	Voiced Ratio	Voiced Seconds	Clipping	Pass?
neutral_p229_002.wav	3.88 s	22.4%	0.86 s	None	PASS

Generation Gate

Model	Total Windows	Passed	Pass Rate	Status
Chatterbox	19	19	100.0%	PASS
XTTS-v2	21	21	100.0%	PASS
Kokoro	20	20	100.0%	PASS

XTTS-v2 had 3 clipped windows (at 60–75s, 90–105s, 225–240s) — documented but not blocking. All windows had adequate voicing.

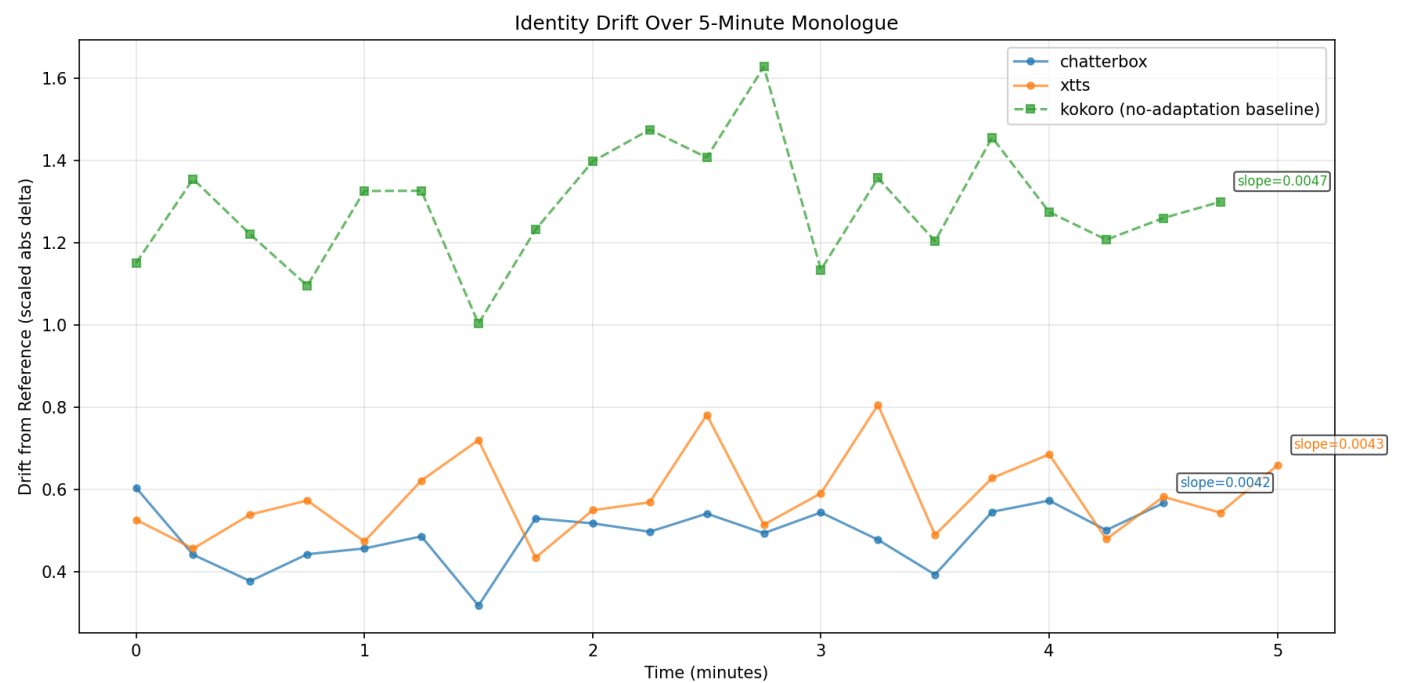
Drift Results

The main drift summary table. Lower drift = closer to reference voice identity.

Model	Drift Mean	Drift Std	Drift Early	Drift Late	Drift Increase	Drift Slope	Raw Variance	Relative Drift (lower = better)
Chatterbox	0.491	0.072	0.467	0.547	0.080	0.00422	—	
XTTS-v2	0.583	0.102	0.514	0.590	0.076	0.00431	—	
Kokoro (no-adaptation)	1.290	0.145	1.229	1.300	0.070	0.00473	13.384	

Bar column scaled to maximum drift mean (1.290 = 100%). Chatterbox (gold) = best, closest to reference.

Drift Over Time



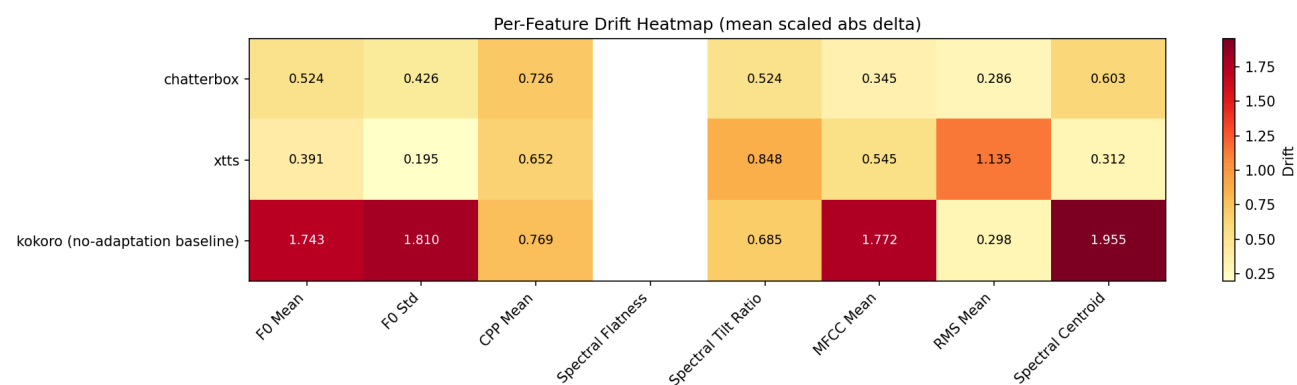
All models show positive drift. Drift rates are very similar (~0.004/window). Chatterbox starts closest but drifts fastest.

Per-Feature Drift

Drift mean broken down by acoustic feature channel. Reveals *which* aspects of voice identity drift over time.

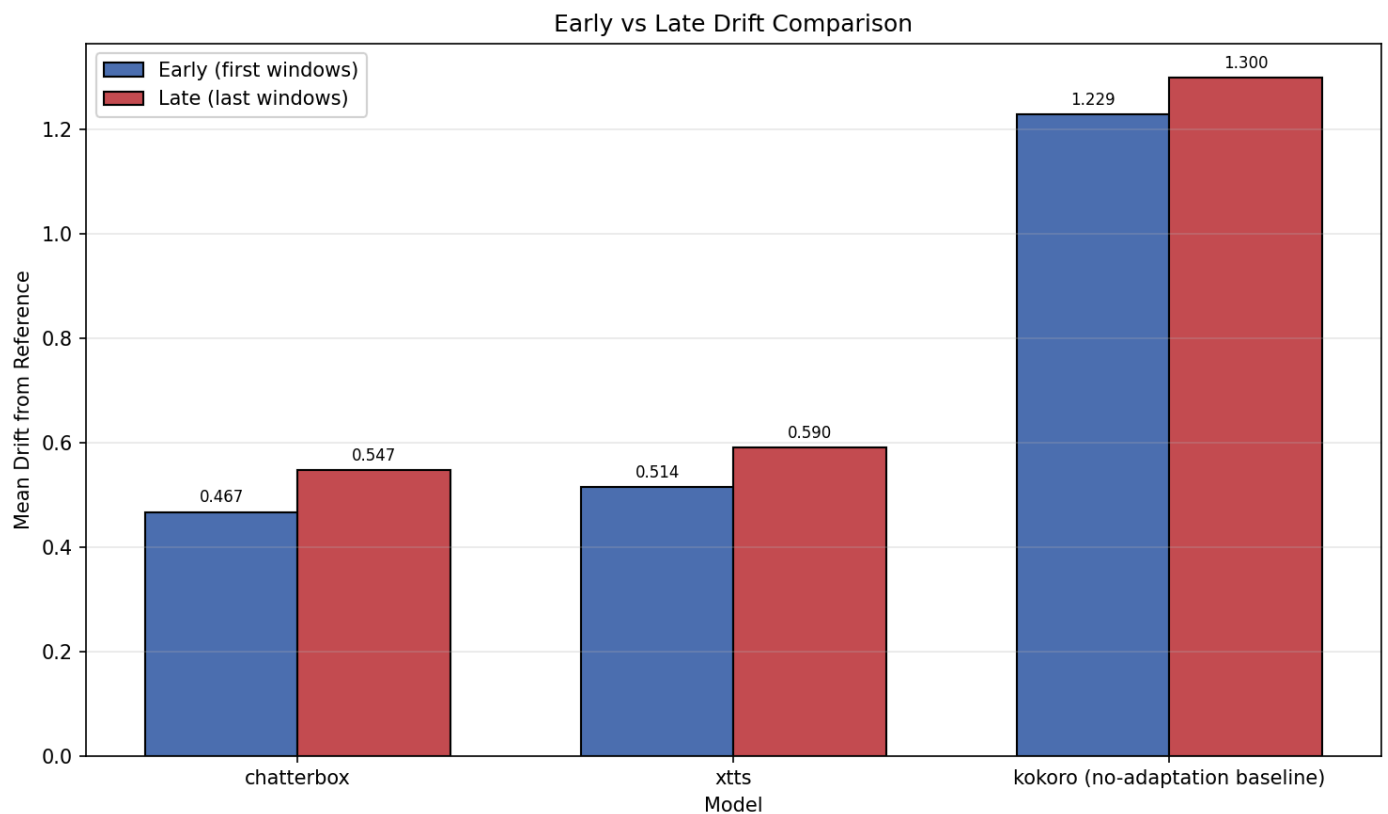
Feature	Chatterbox	XTTS-v2	Kokoro
f0_mean	0.524	0.391	1.743
f0_std	0.426	0.195	1.810
cpp_mean	0.726	0.652	0.769
spectral_tilt	0.524	0.848	0.686
mfcc	0.345	0.545	1.772
rms	0.286	1.135	0.298
centroid	0.603	0.312	1.955

Feature Drift Heatmap



CPP (breathiness) is the highest-drift feature for both voice-cloning models. XTTS-v2 has anomalous RMS drift (1.135), likely from 3 clipped windows.

Early vs Late Drift



Single-Listener Sanity Check

Listener

davilex — 2026-05-25

Question 1: Does Chatterbox change from start to end?

→ **Barely perceptible change.** Something shifts but can't be named. Consistent with CPP drift (0.73) exceeding F0 drift (0.52).

Question 2: Chatterbox end vs XTTS end vs reference?

→ **Both sound similar to reference at 5 minutes.** XTTS no perceptible change from its start. Gap between models (0.09) below perceptual threshold.

Question 3: Kokoro?

→ **Clearly different person (expected).** Most stable within own voice. af_heart is consistently af_heart — confirms no-adaptation baseline works as intended.

Conclusion

5 minutes is too short for perceptible drift. The acoustic instrument is more sensitive than the ear at this timescale. Measurable drift exists but does not cross the perceptual threshold within the tested duration.

Key Findings

Finding 1: Voice identity drifts measurably over 5 minutes across all models

All three models show positive drift — voice moves away from reference over time. Drift increase ranges from +0.070 (Kokoro) to +0.080 (Chatterbox). The effect is systematic and measurable with the 7-feature acoustic instrument.

Finding 2: Chatterbox starts closest, drifts fastest — consistent with autoregressive conditioning decay

Chatterbox has the lowest mean drift (0.491 = best) but the highest drift_increase (+0.080). This pattern — strong initial match followed by gradual divergence — is consistent with autoregressive models losing conditioning signal over long sequences rather than exhibiting random-walk drift.

Finding 3: Nature of drift differs by model

Chatterbox drifts in CPP/breathiness ($\text{drift_cpp} = 0.726$) — the voice loses breathiness texture over time. XTTS-v2 drifts in RMS/loudness ($\text{drift_rms} = 1.135$) — voice gets louder and more variable, likely from clipping artifacts. These are qualitatively different failure modes, not just different magnitudes of the same effect.

Finding 4: CPP (breathiness) is the least stable channel — confirms Paper 2 at long time scale

For both voice-cloning models, CPP drift exceeds F0 drift: Chatterbox $\text{CPP}=0.726$ vs $\text{F0}=0.524$, XTTS $\text{CPP}=0.652$ vs $\text{F0}=0.391$. Breathiness quality degrades faster than pitch under long-form generation — a direct extension of Paper 2's finding that breathiness is the hardest voice characteristic to preserve.

Finding 5: Near-identical drift slopes (~0.004/window) may reflect homogeneous input text

All three models have nearly identical drift slopes: Chatterbox +0.00422, XTTS +0.00431, Kokoro +0.00473. This uniformity is suspicious — it may reflect the structural homogeneity of the input narrative (single paragraph, uniform sentence structure) rather than a fundamental property of long-form TTS generation. Testing with multi-genre text is needed to confirm.

Finding 6: Perceptual threshold not reached at 5 minutes

Acoustic drift is measurable but not reliably audible at this timescale. The 0.092 gap between Chatterbox and XTTS (0.491 vs 0.583) and the 0.080 within-Chatterbox increase are below the perceptual threshold for a single listener. The ear is less sensitive than the acoustic instrument — confirmed by single-listener check.

Connections to Papers 2 & 3

Paper 2 — Breathiness

Paper 2 ranked XTTS-v2 best for short-form breathiness preservation (score 0.097) and Chatterbox second (0.311). In this long-form test, the CPP drift ordering is reversed: XTTS drift_cpp = 0.652 vs Chatterbox's 0.726 — both close.

The breathiness metric (CPP) is the **highest-drift feature** for both voice-cloning models. CPP drift substantially exceeds F0 drift in both cases, confirming that **breathiness quality is the least stable voice characteristic over long generation**.

Praat CPPS function and parameters are identical to Paper 2, enabling direct cross-paper comparison.

Paper 3 — Prosody (F0)

Paper 3 ranked Chatterbox best for F0 contour match (DTW 11.27). In this long-form test, Chatterbox has higher F0 mean drift (0.524) than XTTS (0.391), but XTTS has worse F0 std drift.

The F0 drift is moderate compared to other features — **voice cloning models preserve pitch reasonably well over time**, but breathiness and timbre are less stable. Pitch preservation does not guarantee breathiness preservation.

Same reference speaker (p229) used across all three papers for consistency.

What's Solid

1. **Gate design:** Reference and per-model generation gates passed cleanly, confirming all audio is valid for analysis. No drift measurement rests on unvalidated data.
 2. **Praat CPP alignment:** Breathiness measurement uses the exact function and parameters from Paper 2, enabling direct cross-paper comparison.
 3. **Kokoro baseline framing:** Correctly labeled as no-adaptation baseline in all tables, figures, and analysis. Raw feature variance (13.38) reported separately to distinguish "stable in own voice" from "close to reference."
 4. **Per-window methodology:** Non-overlapping 15-second windows with quality flags, voiced frame ratio filtering, and documented window failures — methodologically sound and reproducible.
 5. **Cross-paper connectivity:** Both breathiness (Paper 2) and prosody (Paper 3) findings are extended to long-form generation, with specific feature-level comparisons that surface the CPP instability finding.
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Limitations

1. Euclidean distance in normalized feature space is a proxy for perceptual voice similarity — not validated against human ratings.
2. No human listening test to confirm whether measured drift is perceptually noticeable (single-listener sanity check only).
3. drift_increase uses only first 5 and last 5 windows (~75 seconds each) — may miss mid-trajectory drift and recovery patterns.
4. 15-second non-overlapping windows may miss sub-window instability; overlapping windows would give smoother trajectories.
5. 5 minutes of generation may be insufficient for significant drift to manifest — longer tests (15–30 min) recommended.
6. Window counts differ between models (19 vs 21 vs 20) — drift_early/late comparisons use different absolute time points.
7. IQR-based robust scaling uses pooled window population (all models combined) rather than reference-only population.
8. Chatterbox generation was chunked (10 chunks of ~100 words) due to CUDA context limitations — chunk boundaries may introduce artifacts.
9. XTTS-v2 had 3 windows with clipping — these windows were included in analysis since they had adequate voicing.
10. Spectral flatness had near-zero variance across all windows, rendering its per-feature drift NaN — contributed minimally to composite scores.
11. Euclidean distance equally weights all 7 features — perceptual importance of each feature channel is unknown.
12. Kokoro drift measures distance to a voice it was never asked to clone (af_heart vs p229).
13. VCTK reference audio (p229) may differ from training data distributions for all models, potentially inflating baseline distances.
14. **Input text homogeneity confound:** The near-identical drift slopes across models (~0.004/window) may partially reflect the uniformity of the single narrative paragraph input text. Real long-form generation faces dialogue, narration shifts, lists, questions, and emotional variation — all of which stress voice consistency differently. Future work should use scripted multi-genre text with controlled prosodic variation.

Output Files

File	Description
results/features/window_features.csv	All features per 15-second window (60 rows)
results/features/reference_features.json	Reference speaker feature anchor
results/features/failed_windows.txt	Windows flagged for clipping or low voicing
results/drift_by_window.csv	Per-window drift scores (60 rows with 7 feature channels)
results/drift_summary.csv	Summary statistics per model
results/gate_check.json	Gate decisions (reference + per-model)
results/figures/drift_over_time.png	Main drift-over-time visualization
results/figures/drift_heatmap.png	Per-feature drift heatmap
results/figures/early_vs_late_drift.png	Early-vs-late stability comparison
results/report.html	This report
data/long_form/{model}/full.wav	Generated long-form audio per model
.gate_passed	Gate sentinel file